Multiphysics simulation enabled '*virtual sensing*' approach for monitoring the parameters in the interelectrode gap during tool-based hybrid laser-electrochemical micromachining

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Abstract

This paper presents a tool-based hybrid laser-electrochemical micromachining process that involves the simultaneous and coaxial application of electrochemical and laser process energies in the same machining zone. In this process, the tool serves both as a multimode waveguide for the laser as well as an ECM tool with sufficient conductive area. The process mechanism involves several multiphysics phenomena occurring in the interelectrode gap (IEG). The IEG is of the order of 20-50 µm. The technological limitations of putting sensors in this small IEG limits monitoring of physical quantities (such as temperature, hydrogen and oxygen gas volume fraction, electrolyte conductivity, current density on the workpiece surface) during the process. This work proposes a 'virtual sensing' approach showing that these physical quantities can be effectively estimated by multiphysics simulations. The evolution of the aforementioned parameters in the IEG is monitored and supported by fundamental explanations. This research is a first step towards making ECM-based processes deterministic and minimizing experimental efforts by development of process digital twins and virtual process-monitoring.

1. Introduction

The term hybrid micromachining refers to a combination of different machining processes/mechanisms to manufacture components with a better

machining performance at sub-millimeter scale [1]. Electrochemical micromachining has been identified as a candidate process for hybridization with other process energies due to processing inherent advantages such as the absence of thermal defects, no tool wear, and independent of workpiece hardness [2]. In prior state of the art, it has been reported that the hybridization of ECM with laser offers improved process localization [3] and gentle material processing [4]. This work presents a tool-based hybrid laser-electrochemical micromachining process that involves the simultaneous and coaxial application of electrochemical and laser process energies in the same machining zone. The process mechanism involves several multiphysics phenomena occurring in the interelectrode gap. The interelectrode gap (IEG) is of the order of 20-50 µm. The technological limitation of putting sensors in this small IEG limits monitoring of physical quantities (such as temperature, hydrogen, and oxygen gas volume fraction, electrolyte conductivity, current density on the workpiece surface) during the process. The electrochemical environment and generation of hydrogen bubbles in and around the IEG further complicate the application of physical sensors. These physical quantities on the other hand can be effectively estimated by multiphysics simulations.

This work proposes a 'virtual sensing' approach enabled by multiphysics simulations for this tool-based laser-ECM process. A multiphysics model is developed in Comsol® which models several phenomena evolving in the hybrid laser-ECM process such as temperature-dependent electrolyte properties, hydrogen and oxygen gas generation, heat generation due to joule heating and laser heating, laminar fluid flow, and electric currents. The properties are evaluated at 5 different (virtual sensor) locations inside the IEG, more specifically on the workpiece surface, tool surface, in the centre of IEG, just at the exit of IEG, and at a specified location on the workpiece surface. The evolution of the aforementioned parameters in the IEG is monitored and supported by fundamental explanations.

2. Tool-based hybrid laser-ECM process scheme

Figure 1(a) shows a process scheme of the tool-based hybrid laser-electrochemical micromachining process which is used in this paper [5]. The main peripherals of this technology are a pulsed voltage source, a pulsed laser source, a hybrid tool, and an electrolyte circulation system. The technology uses a green laser (λ = 532 nm) because of its very low absorption in pure water (Pope 1997). A hybrid tooling concept [6] is used where the tool serves the dual function of an ECM tool-electrode and a waveguide for the laser. The tool is a 1.2 mm outer diameter stainless steel tube with an inner concentric quartz glass hollow capillary. The performance of this waveguide is comparable to a multimode waveguide as the inner diameter of the tool (0.65 mm) is much larger than the wavelength of the laser used (532 nm). This technology makes use of pulsed processes (both laser (ns) and ECM (µs)) to minimize adverse effects such as electrolyte boiling. Figure 1 (b,c,d) shows SEM images of the cavities machined using pure ECM and laser-ECM. In the case of an ECMed cavity, a

central pin is observed below the hollow area of the tool electrode. This is because the current densities at the centre are insufficient for dissolution. However, in the case of hybrid laser-ECM, the central pin is dissolved due to temperature-induced local rise in electrolyte conductivity/enhancement of diffusion of electrolyte species. Here, the shortening or dissolution of the pin is an indicator of the presence of laser-electrochemical removal mechanisms.

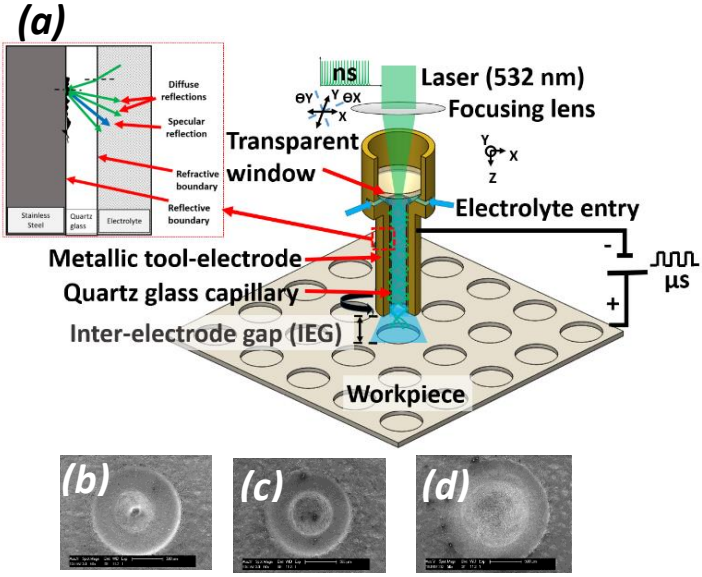

***Figure 1**: (a) Process schematic. SEM images (scale bar 500 µm) of the point removals machined using (b) ECM (c) Hybrid laser-ECM at 36 µJ (measured) (d) Hybrid laser-ECM at 60 µJ (measured). (Constant parameters: 20 V, IEG 20 µm, flow rate 32 ml/min, ECM pulse on-time 10 µs, duty cycle 50 %.)*

3. Model formulation

Figure 2(a) shows a 2D model-approximation of the process at t=0, which is modelled as an axisymmetric model (Figure 2(b)) in Comsol 5.4® software. The model is divided into four domains: domain I, III and IV are solid domains representing Inconel workpiece (IN718), quartz glass capillary, and stainless steel (EN1.4301) tool respectively. The model contains 16 boundaries in total. Figure 2(c) shows the location of 5 virtual sensors along with their coordinates in the IEG. These virtual sensors are implemented in the Comsol® model by employing a dynamic point probe which dynamically probes the model-solution at a defined point in a moving geometry. Figure 3 gives an overview of different physics used in the model of the tool-based hybrid laser-ECM process. The general mesh conditions were: maximum element size 0.539 mm, minimum element size 2.42 µm, maximum element growth rate 1.3, curvature factor 0.3,

resolution of narrow regions 10. The mesh parameters used here are a trade-off between solution time, solution value and automatic remeshing feature. The simulation time was set to 5 s. Table 1 depicts the simulation parameters.

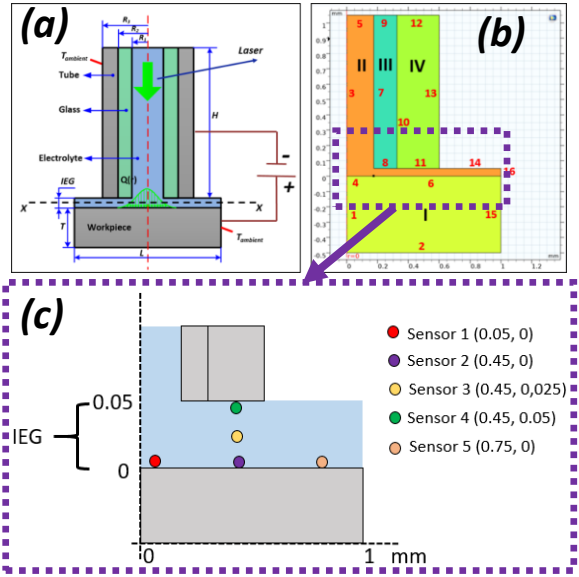

Figure 2: *(a) Schematic of the 2D model (b) 2D axisymmetric model in Comsol (c) A zoom-in view of the IEG showing location of 5 virtual sensors.*

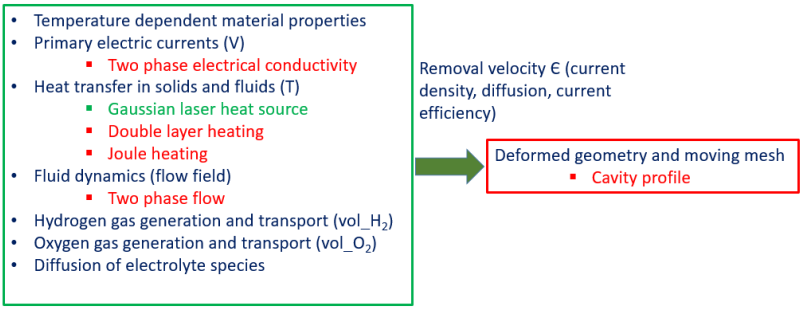

Figure 3: *Overview of the multiphysics model.*

Table 1: *Simulation parameters.*

Parameter	***Value***	***Parameter***	***Value***
Electrolyte conductivity	*12.58 S/m @ 293.15K*	*ECM duty cycle*	*50 %*
Feedrate of tool	*100 µm/min*	V_{oxide}	*2.5 V*
Average molar mass of W/P	*59.79 g/mol*	Q_{flow} *(flow rate at inlet)*	*32 ml/min*
Average valence of W/P	*3.2*	R_{el} *(reflectivity of electrolyte for 532 nm wavelength)*	*0.04*
Density of W/P	*8228 kg/m^3*	α_{IN} *(absorptivity of Inconel for 532 nm wavelength)*	*0.5*
Current efficiency	*Variable (0.845-0.660)*	*Specific material removal (K_IN)*	*1.5063 mm^3/(min.A)*

4. Results and discussion

Figures 4 and 5 depict the data obtained from the five virtual sensors. The variation of the physical quantities (temperature, electrolyte electrical conductivity, current density, diffusion coefficient, volume fraction of hydrogen and oxygen) is probed for a machining duration of 5 s. The observed trends are self-explanatory:

- It can be observed that the laser-induced temperature rise is more pronounced for Sensor 1 due to its location. Consequently, the highest electrolyte conductivity and diffusion coefficient of electrolyte species is also observed at Sensor 1.
- Sensors 2 and 3 record the highest primary current densities as they are directly located below the conductive area of the tool, so in the shortest current-path.
- Sensor 4 is in proximity to the tool and hence shows the highest hydrogen volume fraction followed by Sensor 3.
- Similarly, Sensors 2 and 5 record the highest values of oxygen volume fraction as they are located on the workpiece surface.
- The drop in electrolyte electrical conductivity can be observed at sensors 2 and 4 due to the oxygen and hydrogen gas evolution respectively.

It is also evident that for this laser-ECM process, the temperature effects on physical parameters are very well captured by the model. Specifically, the model sensitivity to temperature changes is observed at all sensors for temperature, electrical conductivity, and diffusion coefficient. The primary current den-

sity shows less sensitivity to laser-ECM as observed from sensor data. However, this effect can be modelled in the future by considering secondary and tertiary current densities.

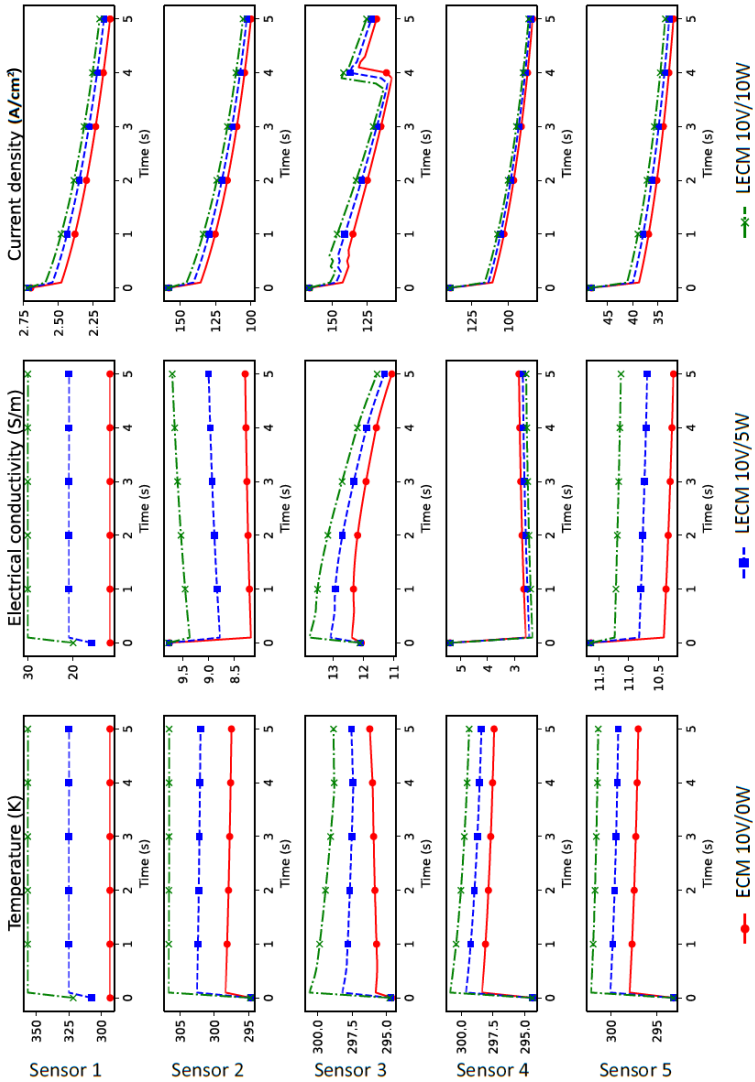

Figure 4: *Dynamic probing of temperature, the electrical conductivity of electrolyte, and current density data at five virtual sensors.*

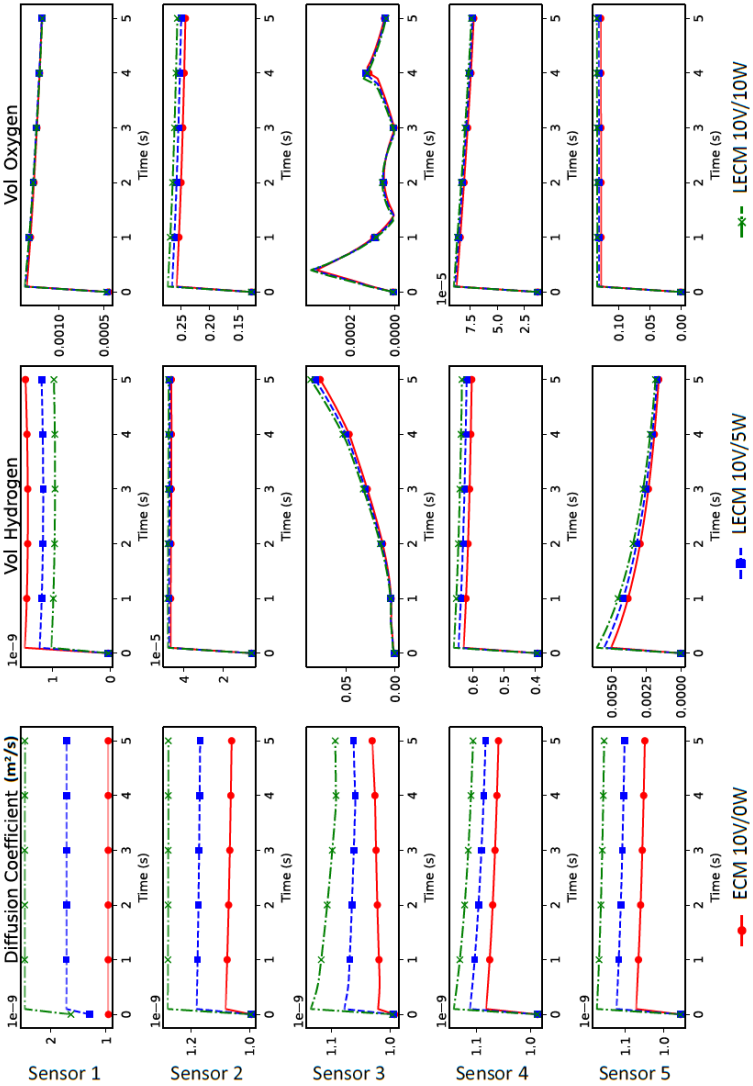

Figure 5: *Dynamic probing of diffusion coefficient, hydrogen volume fraction, and oxygen volume fraction at five virtual sensors.*

5. Concluding remarks

This work presented a 'virtual sensing' approach enabled by multiphysics simulations for monitoring the parameters evolving at different locations in the interelectrode gap during a tool-based hybrid laser-ECM process. This work addresses an important technological limitation of putting an actual sensor in the interelectrode gap and where the 'virtual sensing' approach can be of an added advantage in understanding the process. The model includes several physics inherent to the laser-ECM process on a Comsol® 5.4 platform. The evolution of six different parameters (Electrolyte conductivity, temperature, current density, diffusion coefficient, the volume fraction of oxygen, and hydrogen) has been monitored at five different points (virtual sensors). The temperature sensitivity of the model is visualized in terms of physical quantities inherent to the ECM process. This research is a first step towards making ECM-based processes deterministic and minimizing experimental efforts by the development of the process digital twins and virtual process-monitoring.

Acknowledgments

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